

TrES-3b

R-0038

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_260421 · created 2026-07-08T23:01:30+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461150.4977 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.057
Transit depth (Rp/Rs)^2	3.84 +/- 2.22 [%]
Orbital Inclination (inc)	86.7 +/- 2.87
Airmass coefficient 1 (a1)	0.74 +/- 0.23
Airmass coefficient 2 (a2)	0.18 +/- 0.21
Scatter in the residuals of the lightcurve fit is	37.27 %
Best Comparison Star	#3 - [146, 255]
Optimal Aperture	1.11
Optimal Annulus	4.5
Transit Duration (day)	0.0323 +/- 0.0061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.196 $\pm$ 0.057 vs 0.16309 (deviation 0.58 σ)
Duration fit vs published	0.0323 d vs 0.0567 d (deviation 3.99 σ)
Mid-time O–C	-1879.9 min
Depth SNR	3.4
Residual scatter	37.27 %

Light curve

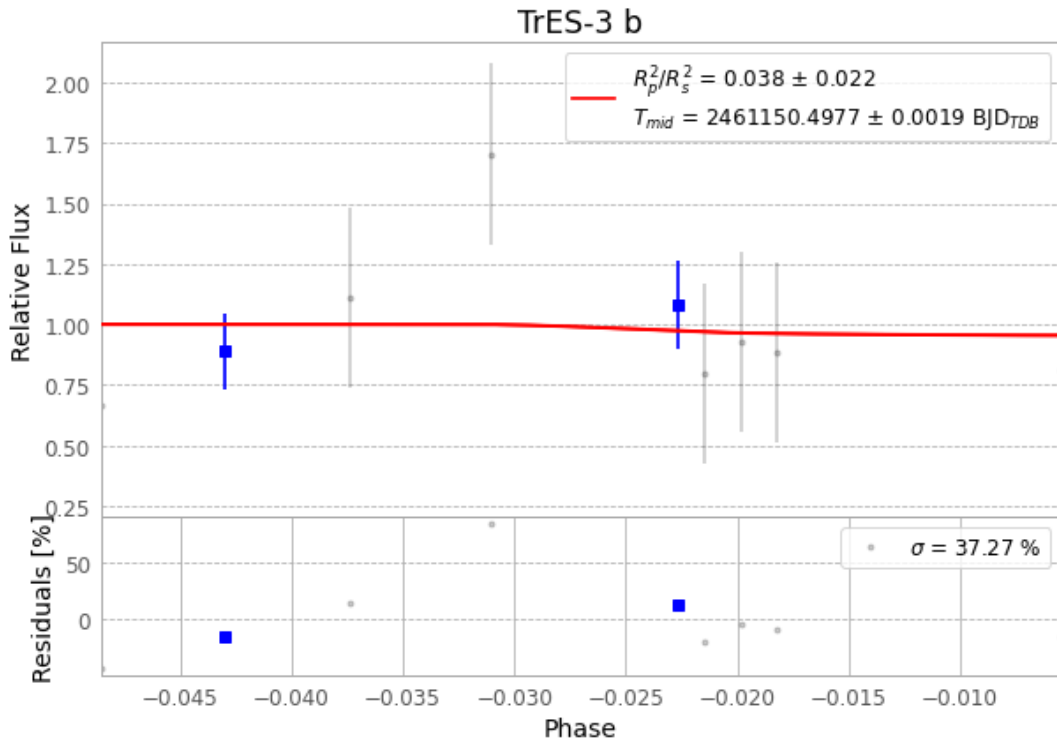


Figure 1 — FinalLightCurve_TrES-3 b_2026-04-20.png (SHA-256 145c3d7a786cac74...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [262, 282], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9e6527cf38bd4314...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

35 FITS frames (23 MB), TRES-3260421053131.FITS → TRES-3260421071216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-260421.log (61bde7d87472...), FinalLightCurve_TrES-3 b_2026-04-20.png (145c3d7a786c...), FinalLightCurve_TrES-3 b_2026-04-20.csv (5e081b3f3b13...)

Compute resources

Wall time 116.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.